

High-quality Color-animated CGH Using a Motor-driven Photomask

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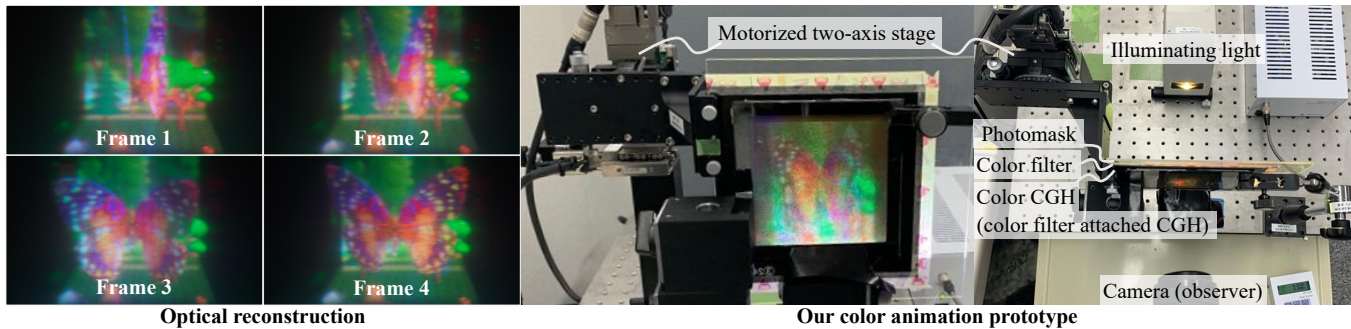


Figure 1: Our color animation prototype and its reconstructed images.

ABSTRACT

We propose a novel display system for a color-animated computer-generated hologram (CGH) with the world's highest level of image quality, screen size, and viewing angle. The conventional method uses a static CGH and pattern illumination to achieve color-animated CGH with large screens and wide viewing angles, but non-illuminated areas result in degraded image quality. Our system solves this problem by using a photomask with a motorized stage, which enables the projection of much finer patterns than with the conventional method. We demonstrated that our method achieves a significant quality improvement compared to the conventional method by developing a color animation prototype with 4 reconstructed images.

CCS CONCEPTS

• **Hardware** → *Displays and imagers.*

KEYWORDS

Computer-generated hologram, 3D display, color animation

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1 INTRODUCTION

Computer holography is a promising 3D display technology that reconstructs realistic 3D images using digital fringe patterns called computer-generated holograms (CGHs). To make it practical, both a large screen size and a wide viewing angle are essential, and these conditions require a CGH with a huge number of pixels and fine pixel pitch. However, color-animated CGH satisfying these requirements has been a longstanding challenge in CGH research.

Several methods using an electronic device called a spatial light modulator (SLM) attempt to solve this challenge. These methods use spatial tiling expansion of reconstructed images [Li et al. 2022; Takaki 2022] and eye-tracking expansion of the viewing angle [An et al. 2020; Häussler et al. 2017]. From another perspective, a near-eye display [Chakravarthula et al. 2022; Kim et al. 2022], which has low requirements regarding screen size and viewing angle, has also been proposed. However, these technologies have problems, such as limiting the number of observers and the inability to achieve a wide viewing area with full parallax.

Recently, we proposed a completely different approach, called color-animated FPHD-CGH [Koiso et al. 2022], to solve the problems above. We first fabricate the static CGH by embedding multiple fringe patterns using two types of space division multiplexing (SDM), one for color (CSDM) and one for animation (ASDM). Then, structured illumination, which turns a beam on and off in the same pattern as ASDM using a laser projector, illuminates the fringe pattern area corresponding to each frame. Erosion is applied to the illuminated areas of this illumination in order to suppress light leakage to other areas. Since the static CGH is fabricated using microfabrication technology that satisfies the above pixel requirements, this method achieves large screen sizes and wide viewing angles. However, this method has a quality problem: non-illuminated CGH regions called dark zones are noticeable as mosaic patterns. This problem arises from the projector's resolution limiting the size

of the illumination pattern. This degradation becomes more pronounced as the number of frames increases, making it impractical to realize an animation of more than two frames. In addition, a dark reconstructed image due to erosion is also undesirable. No previous research has solved these problems.

To solve the above problems, we propose a novel color-animated CGH display system that follows the same basic principles as conventional color-animated FPHD-CGH but does not use structured illumination. Our system uses a photomask driven by a motorized 2-axis stage to control the light illuminating the CGH. The photomask can be produced in very small pattern sizes using lithography technology. Thus, the use of a photomask to project light with a very tiny pattern eliminates the quality degradation and the limitation on animation temporal resolution caused by dark zones. Furthermore, eliminating structured illumination solves dark images caused by erosion and provides robustness in aligning between illuminating light and an ASDM pattern. Our contributions are as follows: (1) We propose a novel color-animated CGH display that eliminates the quality degradation problem and increases temporal resolution; (2) We experimentally determined the optimal pattern size of the photomask; (3) We developed a glasses-free color-animated CGH prototype that successfully demonstrated 4-frame color animation.

2 PROPOSED DISPLAY SYSTEM

2.1 System configuration

The proposed system consists of an illuminating light, a photomask, a motorized two-axis stage, and an SDM-applied color CGH, as shown in Fig. 2. The color CGH is the same as the conventional color-animated FPHD-CGH, which is composed of a CGH with a binary amplitude fringe pattern and a color filter attached to the CGH. The photomask is an opaque plate with a transmission portion that allows light to pass through to form a defined micro-order fine pattern. A fine illumination pattern could be realized by passing the illuminating light through this photomask. We achieved fine illumination pattern switching that far exceeds the lower limit of the conventional structured illumination by designing ASDM as a periodic structure and moving the photomask having the same pattern as ASDM, as shown on the right side in Fig. 2.

As a measure against light leakage at color boundaries, we placed an area without fringe patterns called guard gaps at the boundary of the CSDM, following the conventional color CGH method [Tsuchiyama and Matsushima 2017]. In addition, the transmission area of the photomask is designed to be smaller than ASDM to take into account the stage positioning error to prevent light leakage to the surrounding area, which should be shielded by a mask. Hereinafter, we refer to the area difference between ASDM and the transmission area as the guard area to distinguish it from the guard gap in CSDM, as shown in Fig. 2. The proposed method can suppress light leakage based on the above design, thus, this display system solves the dark zone problem.

2.2 Optical layout

Since the fringe pattern is formed by chromium thin films, the CGH in our system can be used as either a reflection-type or a transmission-type hologram, as shown in Fig. 3. We adopted the

transmission-type hologram because when using the reflection-type hologram, the reflected light on the photomask is an obstacle to viewing the image, as shown in Fig. 3 (a).

There are two types of photomask arrangements for a transmission-type hologram: the illuminating light side (stacked pattern I) and the observer side (stacked pattern II), as shown in Fig. 3 (b). In stacked pattern I, only a specific area of the CGH is illuminated as in the conventional method. On the other hand, in stacked pattern II, the entire CGH is illuminated, and then only the reconstructed light for a specific frame is extracted with a photomask. Theoretically, this difference does not affect image quality because the light reaching the eye is exactly the same in both types. However, the presence of a photomask on the observer's side can interfere with the observation of the reproduced image in practice because the human eye tends to focus on the mask. This effect is extremely limited since we set the mask pattern very finely, but the proposed system adopts stacked pattern I, which is better from a quality perspective. Specifically, the photomask, color filter, and CGH are arranged in this order from the light source side, as shown in Fig. 2.

3 OUR 4-FRAME COLOR-ANIMATED CGH PROTOTYPE AND ITS EVALUATION

3.1 Basic implementation

We implemented a glasses-free prototype of a 4-frame color CGH with a screen size of 104.8576 mm $\times$ 104.8576 mm and a 30-degree viewing angle horizontally and vertically. Note that this CGH can achieve as large size as the conventional color-animated FPHD-CGH because it is fabricated similarly.

The right photographs of Fig. 1 show our prototype, which is composed of a CGH (resolution: 131072 $\times$ 262144, pixel pitch 0.8 μ m $\times$ 0.4 μ m), a color filter (each color stripe 80 μ m with 20 μ m guard gap), Suruga Seiki motorized 2-axis stage (PZG413-R05AG-C, uni-directional positioning accuracy 6 μ m, max speed 10 mm/sec), Sony laser projector (LSPX-P1), and photomask (exposed a silver halide photosensitive film on a PET sheet). We calculate fringe patterns from 3D models corresponding to each frame of the animated CGH using the polygon-based method [Matsushima 2005] and hidden surface removal [Matsushima et al. 2014].

The design of the ASDM for CGH employed a square grid shape so that all frames are adjacent to each other, i.e., all frames could be accessed directly by moving the photomask in a straight line, as shown on the right side of Fig. 2. The width of the guard area of the photomask is set to 12 μ m for vertical only, which is twice the positioning accuracy of the stage. Note that 20 μ m guard gaps are provided by the CSDM in the horizontal direction. Those guard gaps are larger than the guard area's 12 μ m, and also prevent light leakage. Thus, we do not provide guard areas horizontally.

We implemented a virtual 6-frame animation of the butterfly flapping its wings. Since this 6-frame animation is a reciprocating motion, we achieved it with four different frames by using the same frame in duplicate. The one-block sizes of the photomask pattern and ASDM pattern, called the block size, were chosen considering the subjective image quality as described in Section 3.2.

The animation's frame rate depends on the stage's driving speed. Accordingly, the maximum speed of the stage is 10 mm/sec and with a mask with a 240 μ m block size, which was the highest quality

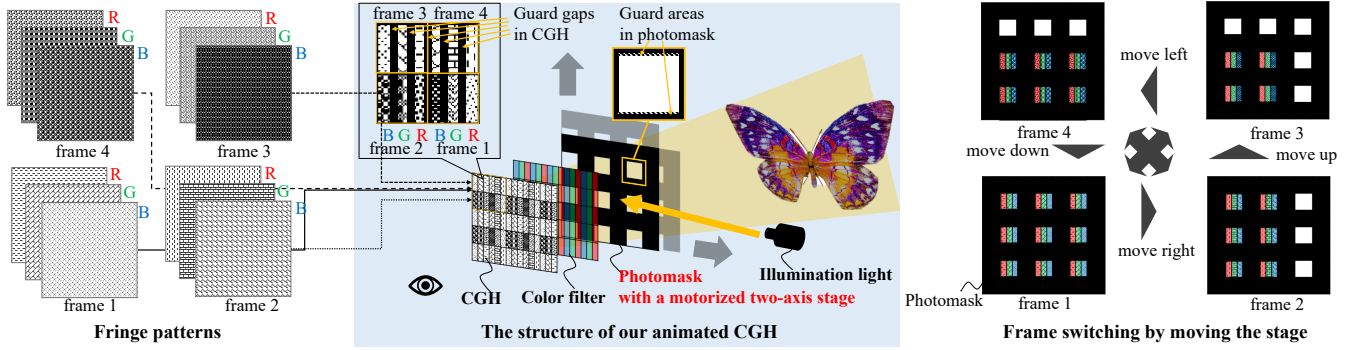


Figure 2: Outline of our proposal

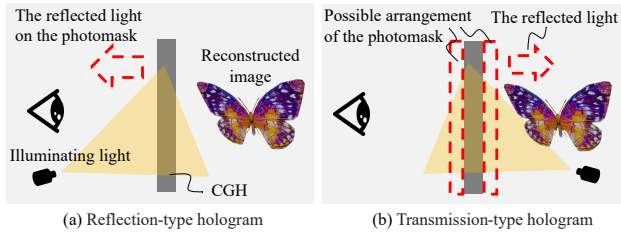


Figure 3: Hologram type

based on the results of the experiment described in the next section, this system can be played back at more than 40 fps.

3.2 Experiments on the optimal block size

We determined the optimal block size for our prototype in terms of image quality through a subjective experiment. Specifically, two photomasks were randomly selected from four photomasks with different block sizes and placed side by side behind the CGH, and their quality was compared while the observers moved their heads left and right and up and down, at an observation distance of 50 cm. This process was conducted for a total of 12 variations, including switching the left and right sides of the photomask's position. The block sizes of the four masks were 80 μm , 160 μm , 240 μm , and 320 μm , respectively, taking 80 μm , the width of the CSDM, as one unit of change so as not to disrupt the color balance of the image.

In this experiment, we used a static color CGH without ASDM. This was to ensure that the conditions affecting the reconstructed image were the same, except for the block size of the photomask. It is acceptable to use a static CGH because it is considered that it simulates the reconstructed image when a single frame of the animated CGH is viewed under optimal conditions.

We asked participants to make a left-right comparison in terms of brightness, noticeability of the photomask, blur, and overall quality and to select the appropriate one from the following options for each evaluation index: better quality on the left, slightly better quality on the left, no difference observed, slightly better quality on the right, and better quality on the right. The best evaluation relative to the comparison target was given a score of 5, and the worst was given a score of 1. We calculated the average value for all participants. A single comparison gives the scores on the left and right masks. We conducted this experiment with a total of 11

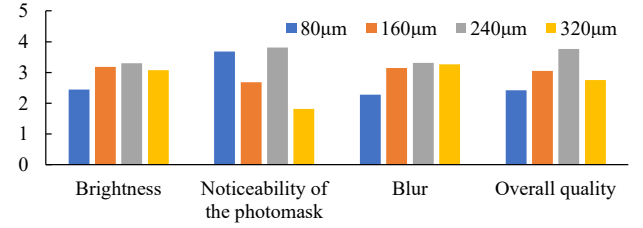


Figure 4: Results of the subjective quality experiment

male and female participants with ages ranging from twenties to fifties.

The results of the experiment are shown in Fig. 4. In terms of the brightness and blur indices, the 80 μm mask significantly underperformed compared to the other mask sizes. In addition, the 160 μm and 320 μm masks gave lower scores for the noticeability of the photomask. The 240 μm photomask, which outperformed the other photomasks for all indices, gave the best overall quality results and was chosen to be the prototype.

3.3 Quality comparison

To verify the effectiveness of the proposed method, we compared it with the conventional color-animated FPHD-CGH. The conventional method is implemented using a scanning laser projector (Sony MP-CLA1) as structured illumination, and the block size of the animated CGH was set to 0.96 mm so that the number of effective pixels per block is approximately 6×6 , similar to conventional research [Koiso et al. 2022]. The optical reconstructions are shown in Fig. 5. All resulting images are captured with a Sony $\alpha 6600$ with fixed camera settings (f-number: 4.5, ISO: 100).

As shown in the first and second rows of Fig. 5¹, the optical reconstruction using the proposed method is brighter, and it is easier to see what is reconstructed compared to that of the conventional method. We also verified that the proposed method solves the problem of dark zones, as shown in Fig. 5 (i) and (f). The results of ours have a slight overlap of the unwanted scaled-up images. This is because we reused the color filter for reflection-type holograms, which take into account the re-entry of light passing through the hologram into the color filter, resulting in light leakage in our transmission-type system without re-entry. This overlap can be solved by making the guard gap larger.

¹Video version is shown in the supplementary video

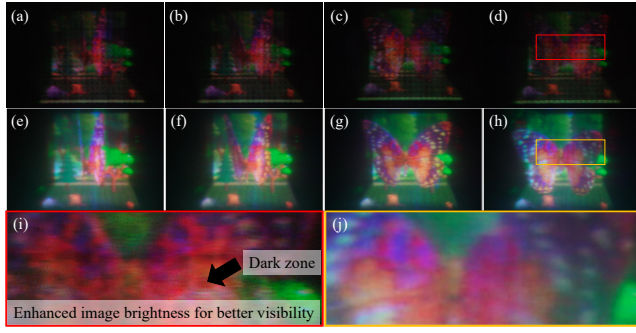


Figure 5: Comparison of optical reconstruction. (a)-(d) conventional, (e)-(h) ours, (i), (j) zoom-in images.

4 DISCUSSION

From the results shown in Fig. 4, we discuss the effect of the block size of the photomask on image quality. For brightness, the smaller the mask is, the greater the magnitude of deteriorated light is due to the guard area, but there was a significant difference only at 80 μm . We consider that this is due to the difference in the area ratios of the guard areas for each mask. Since a guard area is provided for each transmission area, the ratio of the guard area increases as the mask pattern becomes finer. The area ratio of the 12 μm vertical guard area is 15%, 7.5%, 5%, and 3.75% respectively corresponding to block sizes of 80 μm , 160 μm , 240 μm , and 320 μm in each mask, assuming the area ratio of one block as 100%. One notable example from this result is that the 80 μm wide photomask was 7.5% darker than the second darkest mask, and the difference in brightness could be perceived. Considering all the mask sizes, this result indicates that if the difference in the area of transmitted light is less than 2.5%pt, there is no effect on human brightness perception.

We also found that the smaller the block size is, the worse the blurring of the reconstructed image becomes. We consider that this is due to the light diffraction on the photomask. The photomask is a set of rectangular apertures, and the smaller the block size, the more light is diffracted at a larger angle; thus, it causes blurring. In addition, the thickness of the color filter also increases the effect of diffracted light. In our optical arrangement, there is a color filter with a color resist coated on a 0.7 mm glass plate between the CGH and the photomask; thus, diffracted light can spread through the color filter. This problem could be solved by either making the color filter thinner to suppress the effect of light spreading due to diffraction or placing the photomask between the CGH and the color filter. Widening the guard area of the photomask to account for the thickness of the color filter may also be an effective solution. Such implementation issues will be studied in future work.

Regarding the noticeability of the mask, the mask pattern should normally be more conspicuous with a larger block size, but the mask with a block size of 160 μm was considerably more noticeable than the mask with a block size of 240 μm . We consider this to be the effect of the RGB stripe light intensity pattern appearing on the CGH when the 160 μm and 420 μm masks were used rather than the prominence of the mask itself. As shown in Fig. 6 (a)-(d), the pattern period of the color filter and the photomask do not match when using the 160 μm and 320 μm masks. Thus, the amount of RGB light in each small area of the CGH became biased, and this RGB

stripe light pattern was generated. We verified this phenomenon in simulated reconstructions as shown in Fig. 6 (e) and (f).

Based on these considerations, the optimal mask size should be designed so that it is 160 μm or larger in terms of brightness and resolution. Simultaneously, it should be designed to match the period of the color filter in terms of the noticeability of the mask; that is 80 μm or a multiple of 240 μm in this case.

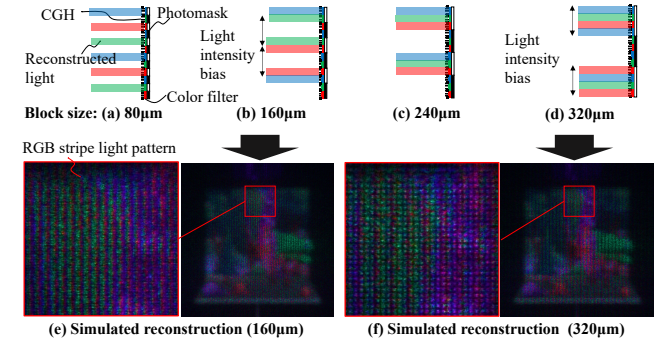


Figure 6: RGB stripes on the CGH surface

5 CONCLUSION

This paper proposes a new display system for color-animated CGH using a photomask and a motorized two-axis stage. Our system solves the conventional method's dark zone problem and the brightness degradation of the reconstructed images. We also experimentally determined the optimal block size of the photomask in terms of quality. Our 4-frame prototype of color-animated CGH using this photomask demonstrated a significant improvement in image quality compared to the conventional method.

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REFERENCES

- J. An, K. Won, Y. Kim, J. Hong, H. Kim, Y. Kim, H. Song, C. Choi, Y. Kim, J. Seo, A. Morozov, H. Park, S. Hong, S. Hwang, K. Kim, and H. Lee. 2020. Slim-panel holographic video display. 11 (2020), 5568.
- P. Chakravarthula, E. Tseng, H. Fuchs, and F. Heide. 2022. Hogel-Free Holography. *ACM Trans. Graph.* 41, 5 (2022).
- R. Häussler, Y. Gritsai, E. Zschau, R. Missbach, H. Sahm, M. Stock, and H. Stolle. 2017. Large real-time holographic 3D displays: enabling components and results. *Appl. Opt.* 56, 13 (2017), F45–F52.
- J. Kim, M. Gopakumar, S. Choi, Y. Peng, W. Lopes, and G. Wetzstein. 2022. Holographic Glasses for Virtual Reality. In *Proceedings of the ACM SIGGRAPH*. 1–8.
- R. Koiso, K. Nonaka, T. Kobayashi, and K. Matsushima. 2022. Color Animated Full-Parallax High-Definition Computer-Generated Hologram. In *SIGGRAPH Asia 2022 Posters (SA '22)*. Association for Computing Machinery, New York, NY, USA.
- J. Li, Q. Smithwick, and D. Chu. 2022. Holobricks: modular coarse integral holographic displays. *Light: Science & Applications* 11 (2022).
- K. Matsushima. 2005. Computer-generated holograms for three-dimensional surface objects with shade and texture. *Appl. Opt.* 44, 22 (2005), 4607–4614.
- K. Matsushima, M. Nakamura, and S. Nakahara. 2014. Silhouette method for hidden surface removal in computer holography and its acceleration using the switch-back technique. *Opt. Express* 22, 20 (2014), 24450–24465.
- Y. Takaki. 2022. Enlargements of Viewing Zone and Screen Size of Holographic Displays Using MEMS SLM Combined with Scanning Systems. *Applied Sciences* 12, 13 (2022).
- Y. Tsuchiyama and K. Matsushima. 2017. Full-color large-scaled computer-generated holograms using RGB color filters. *Opt. Express* 25, 3 (2017), 2016–2030.